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react in a characteristic way, irrespective of the organic molecule in which it is present. In almost all the cases, the reaction of an organic compound takes place at the functional group. They exhibit isomerism which is a unique phenomenon.

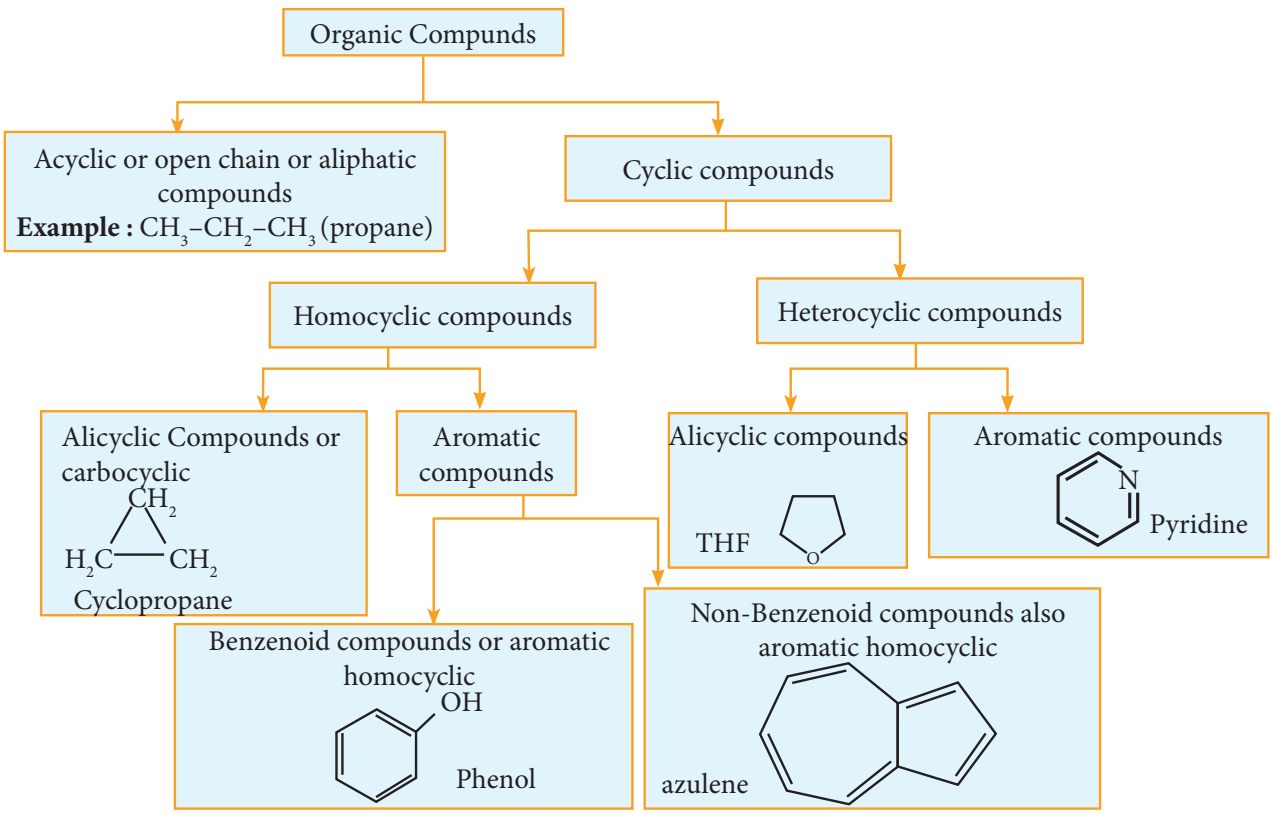
4. **Homologous series:** A series of organic compounds each containing a characteric functional group and the successive members differ from each other in molecular formula by a CH_2 group is called homologous series. Eg.

Alkanes: Methane (CH_4), Ethane (C_2H_6), Propane (C_3H_8) etc..

Alcohols: Methanol (CH_3OH), Ethanol (C_2H_5OH) Propanol (C_3H_7OH) etc..)

Compounds of the homologous series are represented by a general formula Alkanes C_nH_{2n+2}, Alkenes C_nH_{2n}, Alkynes C_nH_{2n-2} and can be prepared by general methods. They show regular gradation in physical properties but have almost similar chemical property.

11.2 Classification of organic compounds

The existing large number of organic compounds and ever-increasing number have made it necessary to classify them. They may be classified based on their structure or the functional group.

11.2.1 Classification based on the structure: